Lecture 21 Tail Bounds

Lecture 21 .pdf
PDF

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Lecture 21 – Tail Bounds

Course: CSE400
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1. Tails

Definition

The **tail of a random variable** X is:

$$P\{X > x\}$$

Why Do We Care About Tails?

Examples of tail events include:

- Jobs delayed more than **24 hours**
- Packets dropped due to **full buffer**

Key Idea

- **Mean → average behavior (easy)**
- **Tail → rare extreme events (hard)**

Why Are Tails Hard to Compute?

Computing tail probabilities requires:

- **Summing many small probabilities**

Thus, tail probabilities often involve large summations over multiple probability terms.

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2. Tails Example 1

Problem

Suppose you are distributing n **jobs among** n **servers at random**.

Question:
What is the probability that a **particular server gets more than** k **jobs**?

Intuition

- Jobs are assigned **randomly**
- Most servers receive **few jobs**
- Occasionally, **one server receives many jobs**

This situation represents a **tail event (rare overload).**

Distribution

Let:

$$X \sim \text{Binomial}(n, p)$$

Then the tail probability is:

$$P(X > k)$$

which involves computing a sum of probabilities of binomial outcomes larger than k.

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3. Tails Example 2

Problem

Jobs arrive to a **datacenter according to a Poisson process** with rate λ jobs/hour.

Question:
What is the probability that **more than** k **jobs arrive during the first hour?**

Intuition

Poisson arrivals in a datacenter:

- Jobs arrive **randomly over time**
- Most hours have **moderate arrivals**
- Occasionally there is a **burst of arrivals**

This represents a **tail event (sudden spike in load).**

Distribution

Let:

$$X \sim \text{Poisson}(\lambda)$$

Tail probability:

$$P\{X > k\}$$

This involves summing probabilities for all values greater than k.

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4. Why Tail Bounds?

Exact tail probabilities are **often hard to compute**.

Examples of exact computations:

$$P\{X > k\} = \sum_{i=k}^{\infty} \cdots$$

or

$$P\{X > k\} = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

These expressions involve:

- **Complicated**
- **Long summations**

Main Idea

Instead of computing the exact value:

$$P(X > k)$$

we compute an **upper bound**.

Approach

Exact:

$$P(X > k) = ?$$

Bound:

$$P(X > k) < \text{something simple}$$

Key Idea

Rather than computing the exact probability:

- Find an **upper bound**
- The bound is **easier to compute**
- The bound still provides a **guarantee**

Thus we **approximate safely instead of computing exactly**.

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5. Running Example (Key Idea)

In the lecture:

- Several **tail bounds** will be developed.
- Each bound will be tested on the **same example**.

Example Distribution

$$X \sim \text{Binomial}(n, \tfrac{1}{2})$$

Experiment

- Flip a **fair coin n times**
- Let:

$$X = \text{number of heads}$$

Mean

$$E[X] = \frac{n}{2}$$

Tail Region

The tail region corresponds to:

- **Unusually many heads**

Question

Compute:

$$P(X > 3n/4)$$

This represents a **rare event far from the mean**.

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6. Markov's Inequality

Key Idea

- Uses **only the mean**
- Large values of a random variable are **rare**

Proof Idea

Consider only the region where:

$$X > a$$

Start from the definition of expectation:

$$E[X] = \sum_x x\, P_X(x)$$

Split the sum into two parts:

$$E[X] = \sum_{x<a} x P_X(x) + \sum_{x\geq a} x P_X(x)$$

Focus on the second term.

Replace x by a (since $x \geq a$):

$$E[X] \geq a \sum_{x\geq a} P_X(x)$$

But

$$\sum_{x\geq a} P_X(x) = P(X \geq a)$$

Thus:

$$E[X] \geq a\, P(X \geq a)$$

Markov Inequality

$$P(X > a) \leq \frac{E[X]}{a}$$

Interpretation

Example intuition:

- If the **average is 10**, then
- Only **few values can be very large (like 100)**

Thus **very large values must be rare**.

Logical Flow of the Lecture

1. Introduced **tail probabilities**

2. Motivated importance using **system examples**
3. Showed examples using **Binomial and Poisson distributions**
4. Explained why **exact computation is difficult**
5. Introduced the idea of **tail bounds**
6. Established **Markov's Inequality** as the first bound using only the mean

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- Turn this into a **clean PDF exam revision sheet**, or
- Convert it into **1–2 page quick revision notes for closed-book exams**.